Recommender Systems
lecture 3: matrix factorization models

Alexey Grishanov

Moscow Institute of Physics and Technology

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Recap: taxonomy

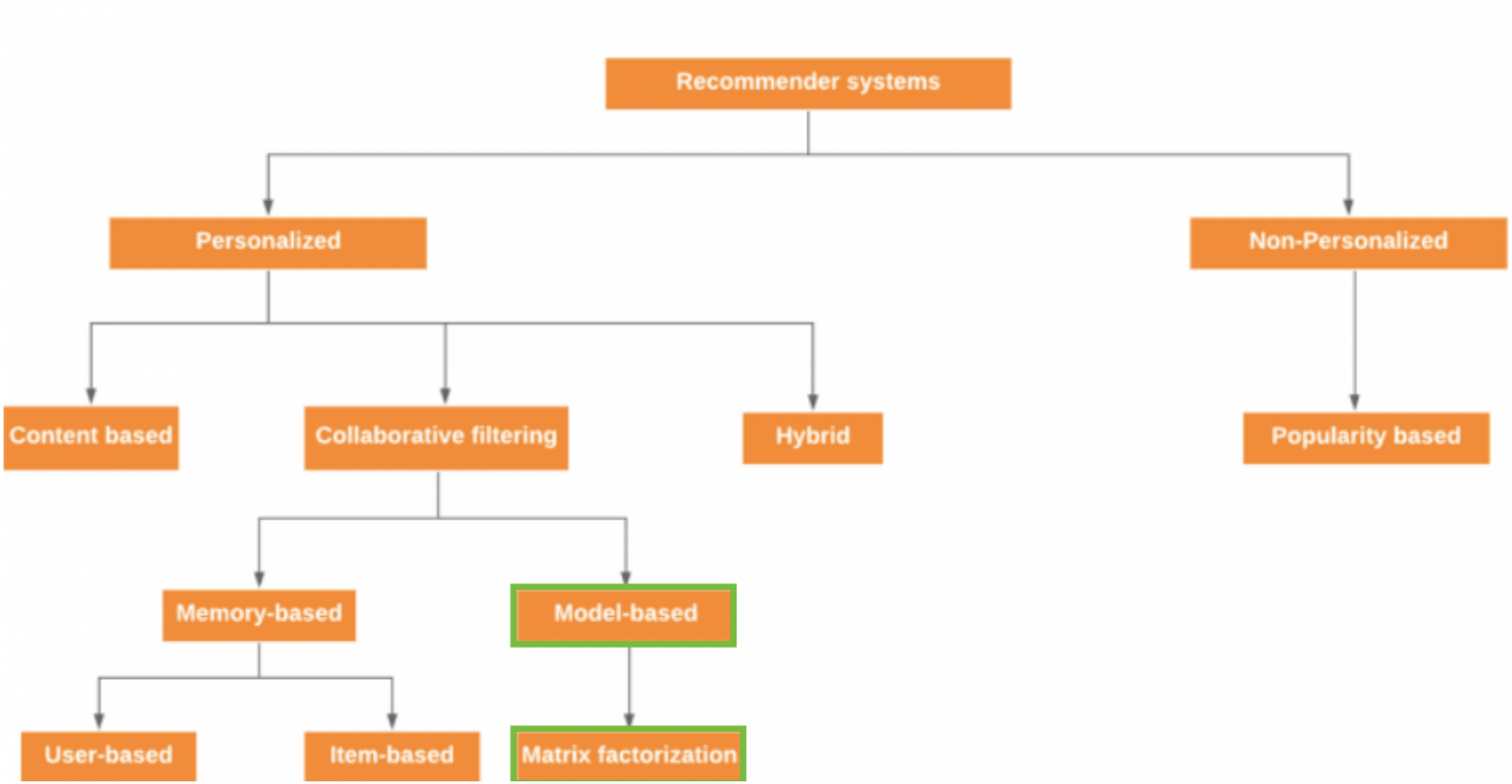


image credit: https://thingsolver.com/introduction-to-recommender-systems

From memory-based to model-based

memory-based

$$r_{ui} = \left[\begin{array}{l} \underset{u' \in U_i}{\text{aggr}}\ r_{u'i} \text{ — user-based} \\ \underset{i' \in I_u}{\text{aggr}}\ r_{ui'} \text{ — item-based} \end{array} \right.$$

cons:

- need to store rating matrix
- struggles in case of lacking intersections

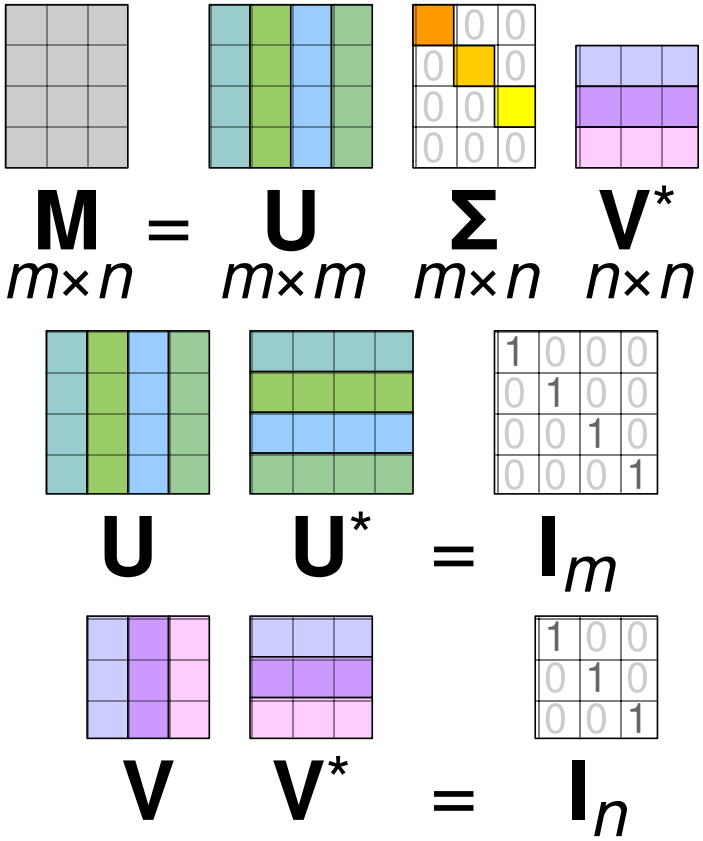
model-based

$$r_{ui} = < p_u, q_i >$$

pros:

- easily scalable
- fast inference

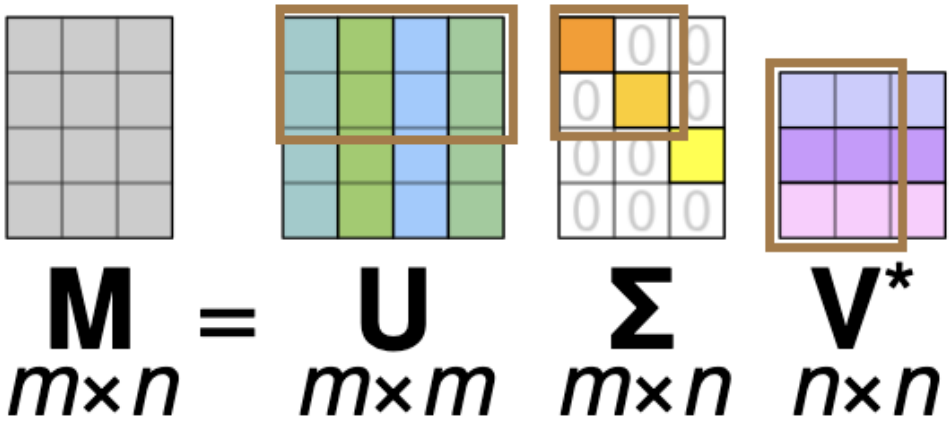
SVD



- Σ — diagonal matrix with non-increasing singular values
- U, V — orthogonal matrices

image credit: https://commons.wikimedia.org/w/index.php?curid=67853297

Truncated SVD



$U_k \Sigma_k V_k^t$ is the optimum of $\|R - B\|_F^2 \to \min\limits_B$

(Eckart–Young–Mirsky theorem)

FunkSVD

idea: instead of computing SVD of partially unobserved R, approximate latent factors using existing interactions

$$\sum_{(i,j)\in R} (r_{ij} - p_i^T q_j)^2 + \sum_{i=1}^{m} \lambda_1 \|p_i\|^2 + \sum_{j=1}^{n} \lambda_2 \|q_j\|^2 \to \min$$

Funk SVD customization

- Adding biases
  $\hat{r}_{ui} = \mu + b_i + b_u + q_i^T p_u$

$$\min_{b_*, q_*, p_*} \sum_{(u,i) \in R} (r_{ui} - \mu - b_i - b_u - q_i^T p_u)^2 + \lambda_4 (b_i^2 + b_u^2 + \|q_i\|^2 + \|p_u\|^2)$$

- SVD++: adding implicit feedback

$$\hat{r}_{ui} = \mu + b_i + b_u + q_i^T \left(p_u + |R(u)|^{-\frac{1}{2}} \sum_{j \in R(u)} y_j \right)$$

  y_j - implicit feedback item vectors,
  q_i - explicit feedback item vector

- timeSVD, timeSVD++

Historical note: SVD on Netflix Prize

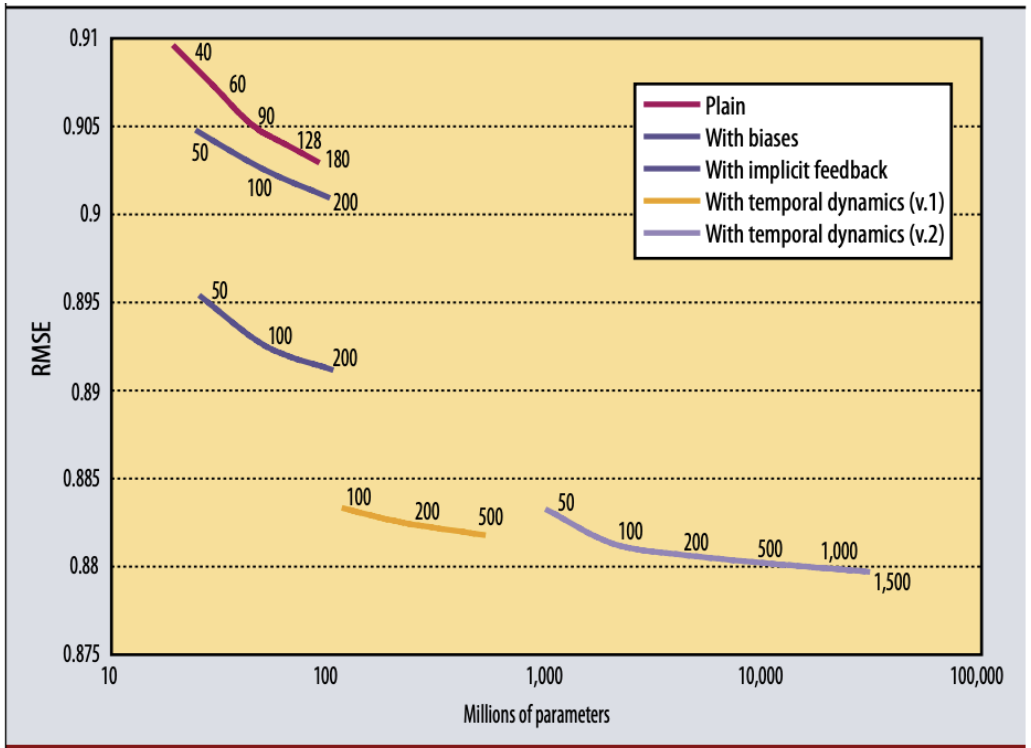


Figure 4. Matrix factorization models' accuracy. The plots show the root-mean-square error of each of four individual factor models (lower is better). Accuracy improves when the factor model's dimensionality (denoted by numbers on the charts) increases. In addition, the more refined factor models, whose descriptions involve more distinct sets of parameters, are more accurate. For comparison, the Netflix system achieves RMSE = 0.9514 on the same dataset, while the grand prize's required accuracy is RMSE = 0.8563.

source:

Matrix factorization techniques for recommender systems, Y. Koren, C. Volinsky

ALS

$$\min_{X,Y} \sum_{r_{ui} \text{is known}} (r_{ui} - x_u^\top y_i)^2 + \lambda \left(\sum_u \|x_u\|^2 + \sum_i \|y_i\|^2 \right)$$

approach:
fix X and optimize Y, fix Y and optimize X, repeat until convergence

ALS algorithm

Algorithm 1 ALS for Matrix Completion

Initialize X, Y
repeat
  for $u = 1 \dots n$ **do**

$$x_u = \Big(\sum_{r_{ui} \in r_{u*}} y_i y_i^\intercal + \lambda I_k \Big)^{-1} \sum_{r_{ui} \in r_{u*}} r_{ui} y_i$$

  end for
  for $i = 1 \dots m$ **do**

$$y_i = \Big(\sum_{r_{ui} \in r_{*i}} x_u x_u^\intercal + \lambda I_k \Big)^{-1} \sum_{r_{ui} \in r_{*i}} r_{ui} x_u$$

  end for
until convergence

source: https://stanford.edu/ rezab/classes/cme323/S15/notes/lec14.pdf

Implicit ALS

$$p_{ui} = \begin{cases} 1 & r_{ui} > 0 \\ 0 & r_{ui} = 0 \end{cases} \qquad c_{ui} = 1 + \alpha r_{ui}$$

$$\min_{x_\star, y_\star} \sum_{u,i} c_{ui}(p_{ui} - x_u^T y_i)^2 + \lambda \left(\sum_u \|x_u\|^2 + \sum_i \|y_i\|^2 \right)$$

$$\Rightarrow x_u = (Y^T C^u Y + \lambda I)^{-1} Y^T C^u p(u)$$

Implicit ALS step, optimized for precomputation

$$x_u = \left(Y^T Y + \lambda I + \sum_{\forall i | p_{ui} \neq 0} (c_{ui} - 1) y_i y_i^T \right)^{-1} \left(\sum_{\forall i | p_{ui} \neq 0} c_{ui} p_{ui} y_i \right)$$

Literature

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2. *Charu C. Aggarwal.* (2016). Recommender Systems: The Textbook.
3. *Simon Funk (the pseudonym of. Brandyn Webb)* (2006) https://sifter.org/ simon/journal/20061211.html
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5. https://benfred.github.io/implicit/
6. *S. Rendle et al.* (2022). Revisiting the Performance of iALS on Item Recommendation Benchmarks